Save South Africa's penguins — give them a home

BOULDERS BEACH, South Africa (AP) — Nesting in the sparkling sand, preening on the rocks and darting through the waters, the penguins on the southern tip of Africa are the ultimate crowd-pleaser. But crisis looms.

Short of food, exposed to predators and the African sun, their numbers are plummeting. But salvation may rest in a simple manmade solution — housing for penguins.

Dotting the shore of this penguin colony near the Cape of Good Hope are 200 nesting boxes, each big enough to house a happy family of parents, eggs and chicks. The experiment has already worked well on a more distant penguin island in South African waters, and wildlife rangers are eager to see whether the boxes recently installed on Boulders Beach, where tourists can watch the birds up close, will prove equally attractive.

"You look at the penguins and think they have a lovely time in sunny South Africa, but it's a struggle," says Monique Ruthenberg, a ranger with the Table Mountain National Park in Cape Town, where summer temperatures recently hit 104 degrees Fahrenheit.

Park authorities installed the boxes — made of a fiberglass mix, shaped like a burrow and dug into the sand to mimic the real nests — at Boulders Beach as part of desperate efforts to protect the dwindling populations of African penguins.

It has been a losing struggle. Numbers of the cute, curious creatures have plummeted from around 3 million in the 1930s to just 120,000 because of overfishing and pollution. Some experts fear the species will die out in as little as a decade, and are particularly alarmed at the prospect of global warming increasing the number of scorching days, raising water temperatures and altering fish migration patterns.

The Boulders Beach colony has fallen 30 percent from a peak of 3,900 birds in 2005 to 2,600 and some of the island colonies have suffered calamitous declines of 50 percent.

The African penguin, also called the jackass because of its bray, is the only one to inhabit the African continent. It has shorter feathers than the Antarctic birds because it doesn't face such cold and is just 20 inches tall.

The Boulders Colony began in 1985 when a couple of penguins moved from a nearby island onto the beach in the naval base of Simon's Town, decided they liked it and stayed. So many followed that authorities had to build fences to prevent them invading people's gardens. But the tourists poured in.

About 600,000 a year now visit Boulders Beach, which boasts that it is the only place in the world where people can swim with penguins. The real life "Happy Feet" are unfazed by all the attention and, apart from a few who were killed while snoozing under visitors' cars, don't seem to have suffered from their contact with humans.

There is a constant risk from pollution. The last big oil spill was in 2000, when 20,000 penguins were trucked about 470 miles from Cape Town to Port Elizabeth to allow workers time to clean up oil from a wrecked tanker while the birds swam home.

But even in years with no big accidents, the South African Foundation for the Conservation of Coastal Birds has to rescue and rehabilitate hundreds of birds whose feathers are covered in oil illegally dumped at sea and washed ashore.

The population fall continues, especially on the more remote Dyer Island where numbers have plummeted from 23,000 breeding pairs in the early 1970s to just 1,500 pairs. Penguins normally mate for life.

"It's horrible," Wilfred Chivell, chairman of the Dyer Island Conservation Trust, who blames bad fishing management for a dwindling supply of sardines and pilchards, the penguins' main food.

Such is the competition for fish that Ruthenberg says young seals attack penguins to rip the fish from their bellies.

Gulls prey on the eggs and young chicks, often working as a team; the nesting penguins leave their eggs to chase away the invaders, while another gull sneaks in behind, she says.

Eggs lie abandoned in the sand because the parents have taken to the water to escape the heat. Once a nesting pair abandons its eggs, other penguins often follow suit.

So volunteers calling themselves the iKapa Honorary Rangers asked the public to sponsor nesting boxes for $20 each. They initially planned 100 boxes but this was doubled thanks to a $2,000 donation from the Species Survival Plan — a cooperation program linking members of the Association of Zoos and Aquariums in the U.S.

The nesting boxes are meant to give the penguins an edge — shelter from the heat and a better defense against egg-stealing gulls — and the 1,000 boxes on the more remote Dyer Island have proven popular, with 80 percent occupancy.

Now Ruthenberg hopes the Boulders Beach penguins that have lost eggs and chicks will learn the lesson and take to the newly installed boxes in time to lay a second batch before the laying season ends in April.[1]

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[1]Nullis, C. 2009. Save South Africa's penguins: Give them a home. Associated Press. March 29. http://addistimes. com/africa-news/1873-save-south-africas-penguinsgive-them-a-home.html

Name:_____period:_____Date:_____

Save South Africa's Penguins

Use text evidence (quotes) to support your ideas.

1) What factors are causing the penguin populations to decline?

2) According to this article what is one direct action we can take to try to slow down the decline of the South African penguin population?

3) Are humans partly responsible for the decline of penguin populations? Why?

4) Do you think renewable energy technologies can help reverse the decline of these penguin populations? Why?

Name:_____ Period:_____Date:_____

	1	2	3	4	5	6	7	8	9
Aluminum Foil									
Bubble Wrap									
Mylar									
Cork Board									
Felt									
Foam Cup									

Each minute, record the temperature for each material, for 9 minutes.

Record the temperature, then subtract minute 1 from minute 9 to get a temperature difference to graph after 9 minutes.
Temperature Differences:

Aluminum Foil- _____

Bubble Wrap-_____

Mylar-_____

Cork Board-_____

Felt-_____

Foam-_____

Directions: Use the temperature difference data to graph temperature differences for each material. (Remember TAILS DRY MIX)

Answer questions based on the graph.

1) Did you see any trends or patterns in the data? **Explain (use data!).**

2) Did color have a significant affect on how well a material conducted or insulated? **Explain (use data!).**

3) What material or combination of materials do you think you will use to construct your penguin house? **Why (use data!)?**

4) What type of **energy transfer** did you observe in this experiment?

5) What types of **energy** did you observe in this experiment?

Igloo Design Packet **Group Names _____**

Give your group a team name _____

Start by answering these questions to think about your design.

Could light get into the igloo and melt the penguin?

Can convection currents rise and fall and enter your igloo?

How is the heat from the black floor of the hot box going to transfer into your igloo?

What are some ways to slow radiation? convection? conduction?

What are some design features of your own house that keep heat out in the summer time that maybe you could adopt into your igloo design (obviously air conditioning is not an option)?

Why did you choose that felt color?

Did you do a test on that material to make sure it works like you want it to?

How does one layer of that material compare to two layers?

If air is such a good insulator, how can you trap more air?

Is it better to have bubbles up or down with bubble wrap?

What is the difference between a heat insulator and a heat conductor?

Background Information: **Heat Insulator**: A material that reduces the rate of heat transfer. **Heat Conductor**: A material that increases the rate of heat transfer. **Conduction**: Conduction is the way thermal energy transfers from one substance to another by direct contact. It can be the direct contact between solids, or between a solid and a fluid. **Kinetic energy** is transferred as the higher temperature atoms or molecules vibrate and collide with cooler atoms or molecules, warming them up and increasing their kinetic energy. "Warmth" is an indication of how much kinetic energy is at the atomic level. **Convection**: Convection occurs when fluids (gases or liquids) sink or rise because the

cooler fluid is denser and sinks. When this happens, the cooler fluid pushes up the warmer fluid and it rises.
Radiation: Radiation is the transfer of energy in the form of electromagnetic waves. Visible light and infrared light are both forms of radiation that transfer heat.

Create your budget for your igloo. You have 100 total credits to spend. Unused points can be added in to your design later.

Penguin Shelter Supply List and Cost

Team Name: _____

Budget: 100 Credits

Supply	Cost per Item	Quantity	Total
Construction Paper (1 sq)	15		
Cotton Ball (1)	10		
Aluminum Foil (1 sq)	15		
Plastic Wrap (1 sq)	15		
Bubble Wrap (1 sq)	20		
Styrofoam Cup (1)	30		
Mylar Sheeting (1 sq)	25		
Cork Board (1 sq)	30		
Felt (1 sq)	25		
Additional Masking Tape 15 cm (1st 30 cm is free)	5		

Total Budget – Total Spent =

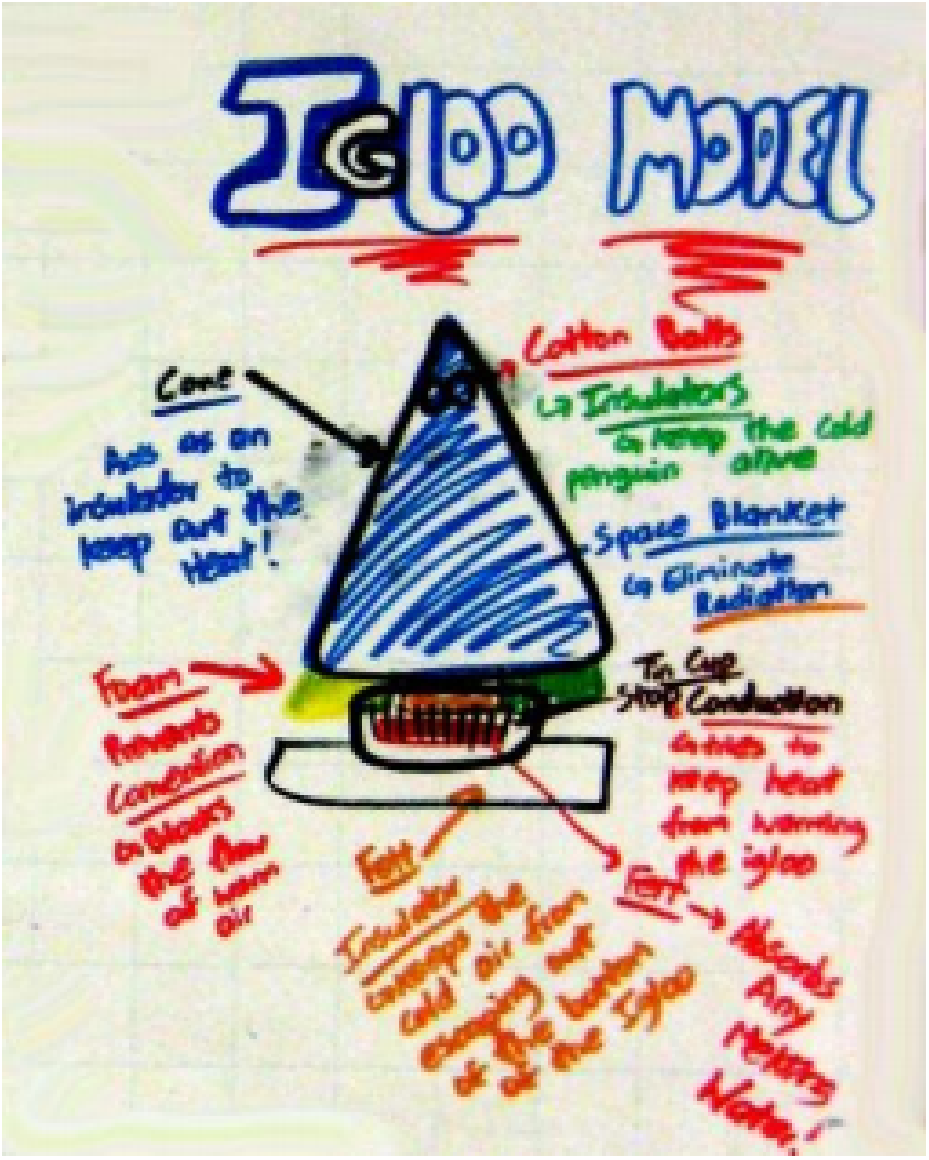
Create a story board for your igloo design.

Include the following things:

1) Draw your design and label the materials you are going to use.

2) Next to each material, indicate the type of heat transfer you are trying to prevent by using that material.

3) Use this example as a guide.



Create <u>your</u> storyboard on the next page:
Igloo Design Storyboard:

Show your completed design packet to your teacher to receive materials and get started.
Receive your teachers initials here: _____

Micro:bit Temperature For Penguin House

Materials:

- 2 Micro:bits
- 2 battery packs

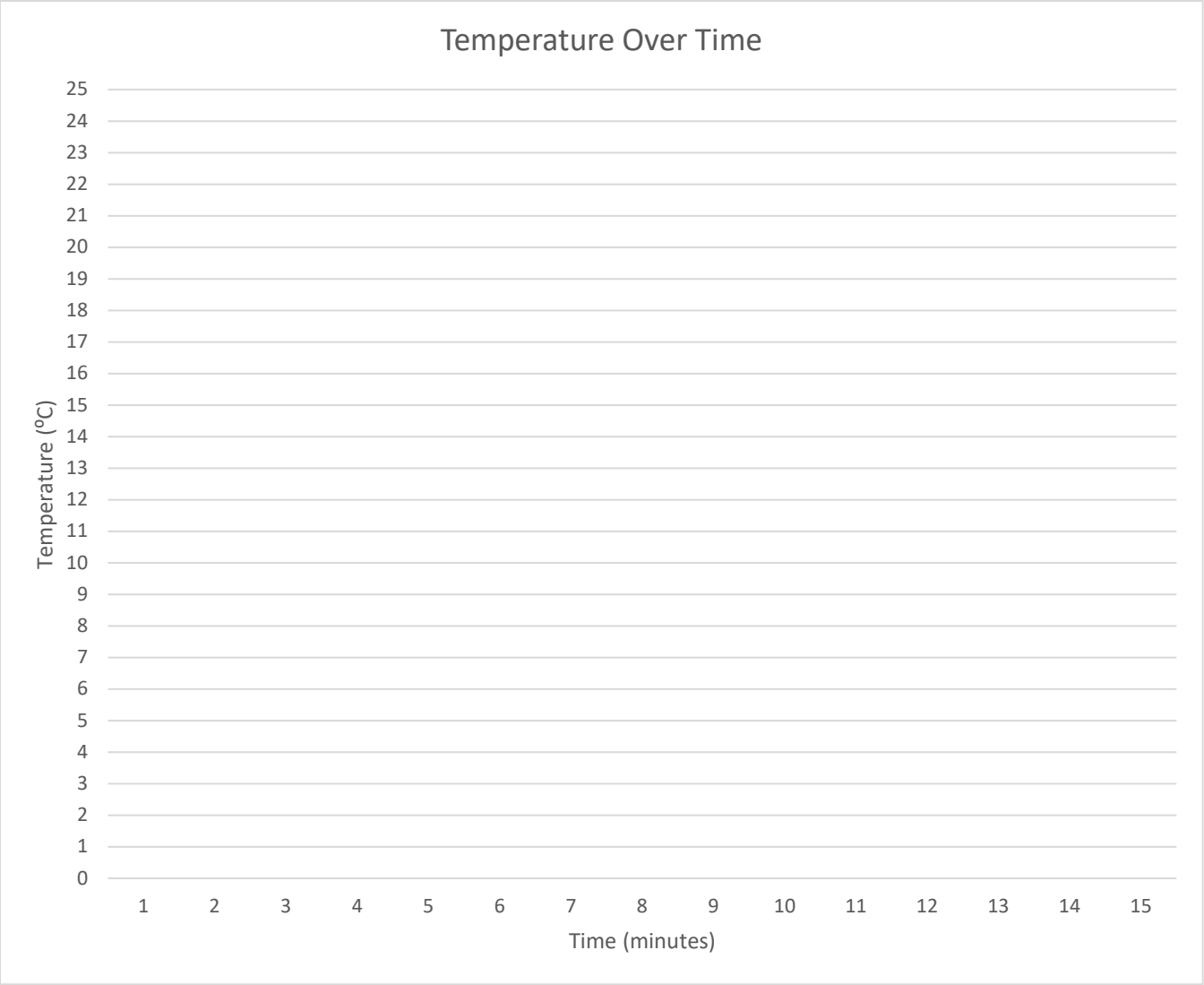
Procedure:

1. Go to makecode.microbit.org
2. Select the variables drawer and create a variable for temp
3. Place a red set variable in the forever block.
4. Go to the input drawer and select the temperature input and place it in the “0” so you set the temp to temperature (^{o}C)
5. Open the Radio drawer and select the “radio send number” and place it on the forever block
6. Then go to variables and select the temp variable and place it in the radio send number block.
7. Go back to the radio drawer and select the block for “on radio received”
8. In the Basic drawer, select the “show number” block and place it in the “on radio received” block
9. Copy the variable for “receivedNumber” and place it in the show number circle
10. Go back to the radio drawer and pull out the “radio set group 1” block
11. Place it in the “on start” block.
12. Connect your micro:bit to the computer.
13. Download your temperature program to both micro:bits.
14. Place one micro:bit in the house for your penguin
15. Record the temperature sent to the other microbit on your table every minute for 15 minutes.

Record your data in the table below. Then, graph your data as a <u>line</u> graph using the graph template on the next page.

Time	Temp	Time	Temp
0 min		8 min	
1 min		9 min	
2 min		10 min	
3 min		11 min	
4 min		12 min	
5 min		13 min	
6 min		14 min	
7 min		15 min	

Temperature Graph



Penguin Igloo Rubric Name _____

Ice Mass Lost During 15 Minute Test	2.5 grams or less (9 pts)	2.6-3.0 grams (8.5 points)	3.1-3.5 grams (8 pts)	3.6-4.0 grams or less (7.5 pts)	4.1-4.5 grams (7 pts)	4.6 grams or more (6 pts)
Micro:Bit Radio and Receiver	Coded and it worked (5 pt)	Didn't work (0)				
Packet Completeness (pages 42-45 in the lilac booklet)	All 4 pages complete (1 pt)	Mostly (75%) complete (.75 pts)	Half (50%) complete (.5 pts)	Less than 50% complete (.25 pts)	Less than 25% complete (0 pts)	

Total Points _____/15